- When the interest is compounded annually, $A = P\left(1+\frac{r}{100}\right)^{n}$.
- When the interest is compounded half yearly, $A = P\left(1+\frac{r}{200}\right)^{2n}$.
- When the interest is compounded quarterly, $A = P\left(1+\frac{r}{400}\right)^{4n}$.
- When the interest is compounded annually but rate of interest differs year by year, $A = P\left(1+\frac{a}{100}\right)\left(1+\frac{b}{100}\right)\left(1+\frac{c}{100}\right)\ldots$ where a, b and c are interest rates for the I, II and III years respectively.
- When interest is compounded annually but time period is in fraction say $a\frac{b}{c}$ years, $A = P\left(1+\frac{r}{100}\right)^{a}\left(1+\frac{\frac{b}{c}\times r}{100}\right)$.
- C.I = A–P (Amount – Principal).
- The simple interest and the compound interest remains the same for the first year or the first conversion period.
- For 2 years, the difference in C.I and S.I is $C.I - S.I = P\left(\frac{r}{100}\right)^{2}$.
- For 3 years, the difference in C.I and S.I is $C.I - S.I = P\left(\frac{r}{100}\right)^{2}\left(3+\frac{r}{100}\right)$.
- x and y are said to vary directly if $\boldsymbol{y=kx}$ always, where k is called the proportionality constant and $k > 0$ assuming that y depends on x and so $\boldsymbol{k=\frac{y}{x}}$.
- x and y are said to vary inversely, if $\boldsymbol{xy = k}$ always, where k is called the proportionality constant and $k > 0$.
- There will be problems which involve a chain of two or more variations in them. This is called as compound variation.
- By finding the proportion, we can use the fact that the product of the extremes is equal to the product of the means to find the unknown (x) in the problem.
- By using the formula $\frac{P_1 \times D_1 \times H_1}{W_1} = \frac{P_2 \times D_2 \times H_2}{W_2}$, we can find the unknown (x).
- We can also find the unknown (x) by Multiplicative Factor Method.
- If two persons X and Y can complete some work individually in $\boldsymbol{a}$ and $\boldsymbol{b}$ days, their one day's work will be $\boldsymbol{\frac{1}{a}}$ and $\boldsymbol{\frac{1}{b}}$ respectively and X and Y together can complete the work in $\boldsymbol{\frac{ab}{a+b}}$ days.

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